

Problem 1

Inverse demand: $P(y) = 300 - \frac{1}{2}y$, equivalently $y(p) = 600 - 2p$.

1(a) Revenues at $y = 200$ and $y = 400$

Revenue is $R(y) = P(y) \cdot y = 300y - \frac{1}{2}y^2$.

$$R(200) = 200 \times 200 = 40,000 \quad R(400) = 100 \times 400 = 40,000$$

Both output levels generate the same revenue, because they lie symmetrically on either side of the revenue maximum ($y = 300$).

1(b) Marginal revenue for $\Delta y = 50$; quantity and price effects

From $y = 200$ to $y = 250$: $R(250) = 175 \times 250 = 43,750$, so $\Delta R = +3,750$ (75 per unit). Decomposition:

$$QE = p_0 \cdot \Delta y = 200 \times 50 = +10,000 \quad PE = \Delta p \cdot y_1 = (-25) \times 250 = -6,250 \quad \Delta R = +3,750$$

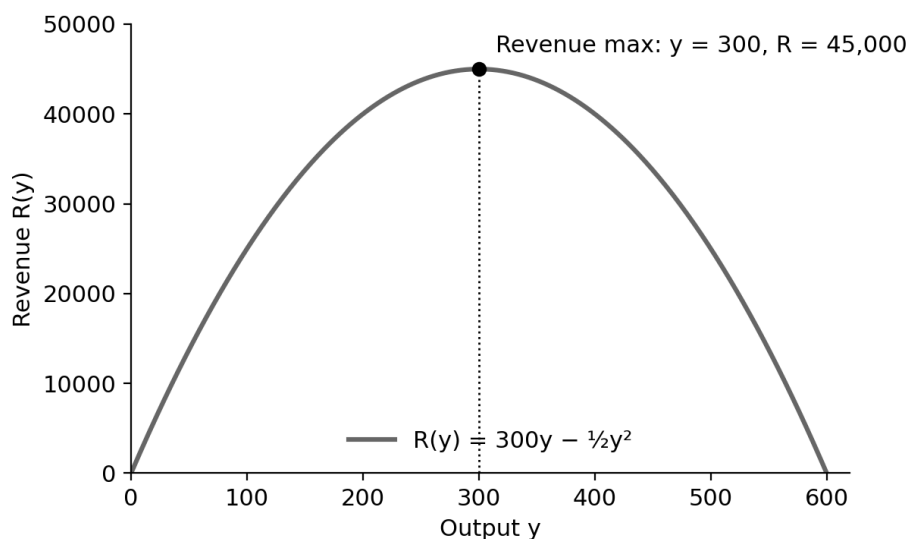
From $y = 400$ to $y = 450$: $R(450) = 75 \times 450 = 33,750$, so $\Delta R = -6,250$ (-125 per unit). Decomposition:

$$QE = 100 \times 50 = +5,000 \quad PE = (-25) \times 450 = -11,250 \quad \Delta R = -6,250$$

Interpretation: selling more units raises revenue (quantity effect) but requires a lower price on all units sold (price effect). In the elastic region ($y = 200$) the quantity effect dominates, so revenue rises; in the inelastic region ($y = 400$) the price effect dominates, so revenue falls.

1(c) Revenue function and revenue maximum

Marginal revenue is $MR(y) = 300 - y$. Setting $MR = 0$ gives $y = 300$, $p = 150$, and maximum revenue $R = 150 \times 300 = 45,000$.

**1(d) Price elasticity of demand at the revenue maximum**

With $y = 600 - 2p$, elasticity is $\varepsilon = (dy/dp)(p/y) = -2p/y$. At $p = 150$, $y = 300$:

$$\varepsilon = -2 \times 150 / 300 = -1$$

Revenue is maximized exactly where demand is unit-elastic ($|\varepsilon| = 1$).

1(e) Supply with $MC = FC = 0$, and with $MC = c$

With zero costs, profit maximization coincides with revenue maximization: $y^* = 300$, $p^* = 150$, and $\epsilon = -1$ in the optimum. With constant marginal costs $0 < c < 300$, the condition $MR = MC$ gives $300 - y = c$, hence:

$$y^* = 300 - c \quad p^* = 150 + c/2 \quad \epsilon^* = -(300 + c)/(300 - c)$$

As c rises, output falls, price rises, and the monopolist operates deeper in the elastic region of demand ($|\epsilon^*| > 1$). A monopolist never produces in the inelastic region.

2(a) Profit-maximizing supply with $c(y) = FC + 100y$

For $y > 0$, $MC = 100$. $MR = MC$ gives $300 - y = 100$, so $y^* = 200$ and $p^* = 300 - 100 = 200$ — provided the resulting profit covers fixed costs. Profit is $\pi = (200 - 100) \times 200 - FC = 20,000 - FC$, so the monopolist produces $y^* = 200$ if $FC \leq 20,000$, and shuts down ($y^* = 0$) otherwise.

2(b) $FC = 10,000$

Since $FC = 10,000 \leq 20,000$, the monopolist produces **$y^* = 200$ at $p^* = 200$** , earning $\pi = 20,000 - 10,000 = 10,000$.

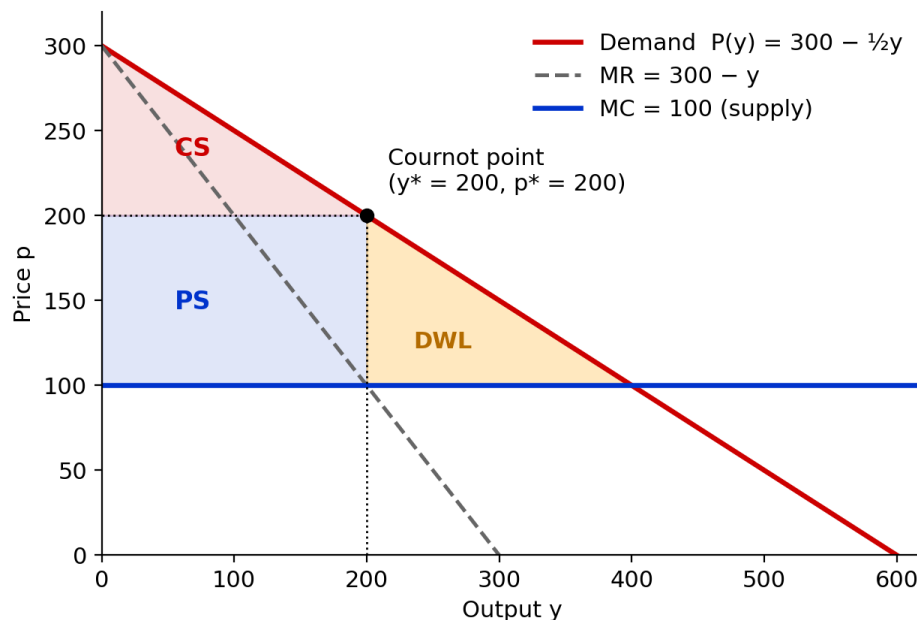
2(c) CS, PS, and deadweight loss

$$CS = \frac{1}{2} \times (300 - 200) \times 200 = 10,000$$

$$PS = (p^* - MC) \times y^* = (200 - 100) \times 200 = 20,000 \quad CS + PS = 30,000$$

Under perfect competition, $p = MC = 100$ and $y = 400$, giving total surplus $\frac{1}{2} \times (300 - 100) \times 400 = 40,000$. The deadweight loss of the monopoly is therefore:

$$DWL = 40,000 - 30,000 = \frac{1}{2} \times (200 - 100) \times (400 - 200) = 10,000$$



Problem 2

Marginal willingness to pay: $P_L(x) = 5 - x$ (L-type) and $P_H(x) = 8 - x$ (H-type); one individual of each type; $MC = FC = 0$.

2.1 Total willingness to pay

Total WTP is the area under the marginal WTP curve:

$$W_L(x) = 5x - \frac{1}{2}x^2 \text{ (for } x \leq 5) \quad W_H(x) = 8x - \frac{1}{2}x^2 \text{ (for } x \leq 8)$$

2.2 Observable types (first-degree price discrimination)

If the monopolist can identify each type, she sells each the Pareto-efficient quantity — where marginal WTP equals $MC = 0$ — and extracts the entire WTP as the price:

$$x_L^* = 5, \text{ price} = W_L(5) = 12.5 \quad x_H^* = 8, \text{ price} = W_H(8) = 32$$

Profit = 12.5 + 32 = 44.5; CS = 0. The allocation is Pareto-efficient: every unit valued above marginal cost is produced, so total surplus is maximized — it just accrues entirely to the monopolist.

2.3(a) Single contract accepted by the H-type

The best single bundle $\{x, p\}$ aimed only at the H-type maximizes $p = W_H(x)$, which is increasing up to $x = 8$. Hence **$\{x, p\} = \{8, 32\}$** . The L-type declines (W_L is at most $12.5 < 32$). Profit = 32; CS = 0.

2.3(b) Single contract accepted by the L-type

If the L-type accepts, the H-type accepts too ($W_H > W_L$ everywhere). The price can be at most $W_L(x)$, so the monopolist maximizes $2 \cdot W_L(x)$, which yields $x = 5$ and $p = W_L(5) = 12.5$, i.e. **$\{x, p\} = \{5, 12.5\}$** . Profit = 25. The L-type keeps zero surplus; the H-type keeps $W_H(5) - 12.5 = 27.5 - 12.5 = 15$. CS = 15.

2.3(c) Two contracts with Pareto-efficient quantities

Quantities are $x_L = 5$ and $x_H = 8$. The L-contract extracts the full WTP: $p_L = W_L(5) = 12.5$. The H-contract must be incentive-compatible — the H-type must prefer it to taking the L-contract:

$$W_H(8) - p_H \geq W_H(5) - p_L \Rightarrow 32 - p_H \geq 27.5 - 12.5 = 15 \Rightarrow p_H = 17$$

Contracts: $\{5, 12.5\}$ and $\{8, 17\}$. Profit = 29.5; CS = 15 (the H-type's information rent). The monopolist earns less than under 3(a)+full extraction because she cannot tell the types apart.

2.3(d) Distorting the L-quantity: $x_L = 2$

Now $p_L = W_L(2) = 5 \times 2 - \frac{1}{2} \times 4 = 8$. The H-type's outside option shrinks: $W_H(2) - 8 = 14 - 8 = 6$, so incentive compatibility allows $p_H = 32 - 6 = 26$.

Contracts: $\{2, 8\}$ and $\{8, 26\}$. Profit = 34 (up from 29.5); CS = 6 (down from 15). Reducing the quantity offered to the L-type makes the L-contract less attractive to the H-type, cutting the H-type's information rent by more than the revenue lost on the L-contract.

2.3(e) Profit-maximizing menu (second-degree price discrimination)

Keeping $x_H = 8$ (no distortion at the top) and choosing x_L to maximize $\pi(x_L) = W_L(x_L) + [W_H(8) - (W_H(x_L) - W_L(x_L))] = 2W_L(x_L) - W_H(x_L) + 32$:

$$d\pi/dx_L = 2(5 - x_L) - (8 - x_L) = 2 - x_L = 0 \Rightarrow x_L^* = 2$$

So the optimal menu is exactly the one in 3(d): **$\{2, 8\}$ for the L-type and $\{8, 26\}$ for the H-type, profit = 34, CS = 6.** Serving only the H-type would yield $32 < 34$, so serving both is optimal. The classic screening

features appear: efficient quantity at the top, downward-distorted quantity at the bottom, and an information rent for the high type.

Problem 3

Demands: $x_1(p_1) = 300 - 2p_1$ and $x_2(p_2) = 200 - 2p_2$; constant marginal cost $c > 0$; third-degree price discrimination.

3.1 Optimal prices and quantities

With separable markets, the monopolist sets $MR = c$ in each market. Inverse demands: $p_1 = 150 - \frac{1}{2}x_1$ and $p_2 = 100 - \frac{1}{2}x_2$, so $MR_1 = 150 - x_1$ and $MR_2 = 100 - x_2$:

$$x_1^* = 150 - c, \quad p_1^* = 75 + c/2 \quad x_2^* = 100 - c, \quad p_2^* = 50 + c/2$$

3.2 Condition for supplying both markets

Market 2 is served only if $x_2^* > 0$, i.e. **$c < 100$** . (Market 1 requires $c < 150$.)

3.3 $c = 50$: prices, quantities, elasticities

$$p_1^* = 100, \quad x_1^* = 100 \quad p_2^* = 75, \quad x_2^* = 50$$

Elasticities $\epsilon = (dx/dp)(p/x) = -2p/x$ at the optimum:

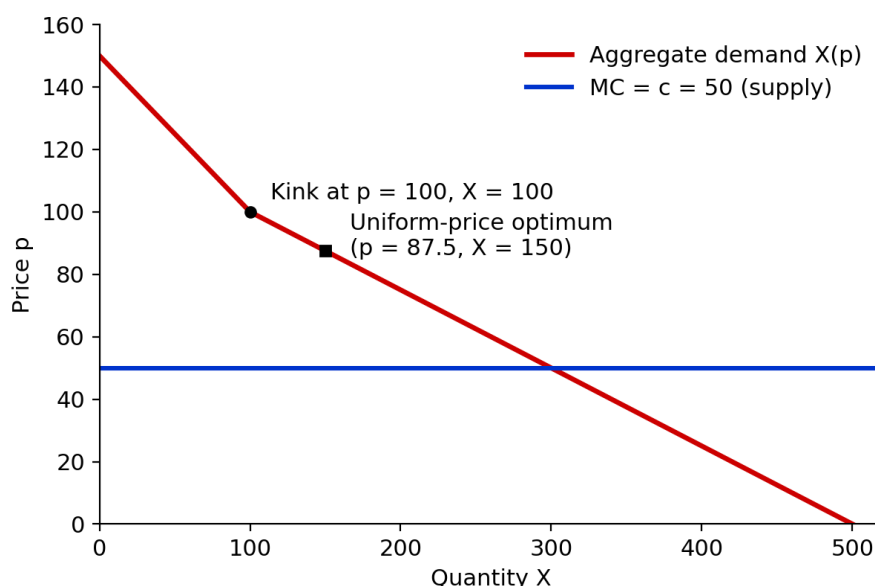
$$\epsilon_1 = -2 \times 100/100 = -2 \quad \epsilon_2 = -2 \times 75/50 = -3$$

The inverse-elasticity (Amoroso–Robinson) rule holds: $(p - c)/p = -1/\epsilon$. Market 1: $50/100 = \frac{1}{2}$; market 2: $25/75 = \frac{1}{3}$. **The more elastic market gets the lower price** — consumers who react more strongly to price changes are charged less.

3.4 Profit function without discrimination; aggregate demand

With a uniform price, aggregate demand is kinked: for $100 \leq p \leq 150$ only market 1 buys, $X(p) = 300 - 2p$; for $p < 100$ both buy, $X(p) = 500 - 4p$. The profit function is correspondingly piecewise:

$$\pi(p) = (p - c)(300 - 2p) \text{ for } 100 \leq p \leq 150 \quad \pi(p) = (p - c)(500 - 4p) \text{ for } p < 100$$



3.5 Optimal uniform price; condition to serve only market 1

If both markets are served, maximizing $(p - c)(500 - 4p)$ gives $p = 62.5 + c/2$ and $X = 250 - 2c$, with profit $(125 - c)^2$. If only market 1 is served, the optimum from 3.1 applies at the constraint $p \geq 100$, with profit $(150 - c)^2/2$.

For $c = 50$: $p = 87.5$, $X = 150$ ($x_1 = 125$, $x_2 = 25$), $\pi = 5,625$ — better than serving only market 1 ($\pi = 5,000$). In general, serving only market 1 is optimal when $(150 - c)^2/2 > (125 - c)^2$, i.e. when

$$c > 100 - 25\sqrt{2} \approx 64.6 \quad (\text{and } c < 150)$$

For high enough marginal costs the low-WTP market is not worth the price cut it forces on the high-WTP market, so the monopolist prices it out.

Problem 4

A monopoly requires a single seller in the relevant market — the product and its close substitutes. Branding alone does not create a monopoly.

1. Shell — diesel: no monopoly. Diesel is a largely homogeneous good sold by many competing station networks; consumers can switch at negligible cost. Shell is one supplier among many in a competitive (at most oligopolistic) market.

2. Swiss Post — letters below 50 g: monopoly. This is a statutory (legal) monopoly: Swiss postal law reserves the delivery of letters up to 50 g to Swiss Post. Entry is prohibited by regulation, so no close substitute supplier exists within the reserved segment.

3. Airbus — Airbus civil aircraft: no monopoly in the relevant market. Airbus is trivially the only producer of aircraft branded 'Airbus', but the relevant market is large civil aircraft, where Boeing (and increasingly other manufacturers such as Embraer or COMAC in some segments) offers close substitutes. The market is best described as a duopoly/oligopoly.

4. Red Bull — Red Bull energy drink: no monopoly. The trademark gives Red Bull exclusivity over the brand, but the relevant market is energy drinks (Monster, Rockstar, private labels, and arguably coffee/soft drinks). Strong branding gives some price-setting power — monopolistic competition — but not a monopoly.

Problem 5

5.1 Choosing a pricing technique

Intuitive pricing: prices are set on experience, rules of thumb, or gut feeling (e.g., cost-plus markups, matching competitors). Cheap and fast, but not grounded in measured willingness to pay, so it can leave large profits on the table.

Revealed preferences: willingness to pay is inferred from actual purchasing behavior — historical sales at different prices, regional price variation, or controlled pricing experiments. Behavior is incentive-compatible (people 'vote' with real money), so estimates are reliable, but it requires data and price variation, and only covers situations similar to those observed.

Stated preferences: customers are asked directly, via surveys or conjoint analysis, what they would pay. Flexible (works for new products and hypothetical changes), but answers are hypothetical and subject to strategic and hypothetical bias.

For an established soda brand with rich transaction data, **revealed preferences** is the most suitable primary technique: demand can be estimated from real behavior, ideally sharpened with small pricing experiments across regions or channels. Stated-preference surveys are a useful complement for new products or packaging, and intuition remains a sanity check — but not the basis of the strategy.

5.2 Should prices rise because purchasing power rose?

Not necessarily — the boss's reasoning skips a step. The optimal monopoly price depends on the demand curve for our soda, in particular its price elasticity (via $MR = MC$ and the inverse-elasticity rule), not on income directly. Higher purchasing power raises the optimal price only if it shifts demand for our product outward or makes it less elastic. For soda that is doubtful: income elasticity of mainstream soft drinks is low, and richer customers may trade up to premium alternatives, making our demand *more* elastic. The right response is to estimate how demand actually changed — using the revealed-preference tools above — and adjust price only if the data show higher willingness to pay. So: partially agree with the motivation, disagree with the automatic conclusion.

Problem 6

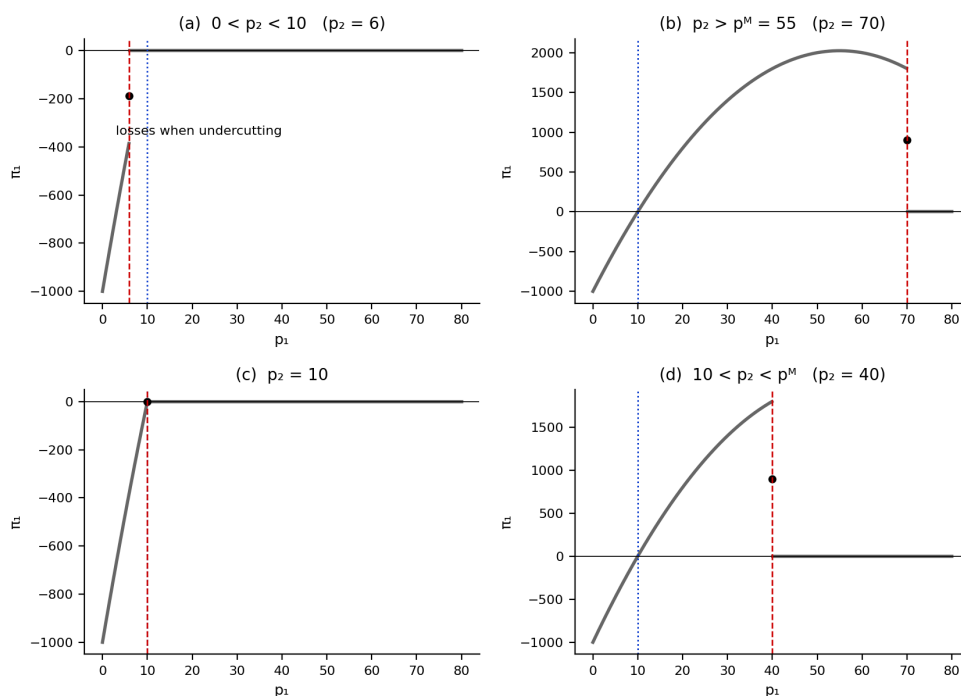
Demand $X(p) = 100 - p$; two firms, $MC = c = 10$, no fixed costs, Bertrand price competition; equal split at equal prices.

6.1 Firm-specific demand

$$x_1(p_1, p_2) = 100 - p_1 \text{ if } p_1 < p_2; \quad \frac{1}{2}(100 - p_1) \text{ if } p_1 = p_2; \quad 0 \text{ if } p_1 > p_2$$

6.2 Firm 1's profit function under four conditions

Monopoly price: maximize $(p - 10)(100 - p) \Rightarrow p^M = 55$.



(a) $0 < p_2 < 10$: undercutting means selling below marginal cost — losses; any $p_1 > p_2$ yields zero. Best attainable profit is 0.

(b) $p_2 > p^M$: firm 1 can charge the monopoly price without being undercut: π_1 follows the monopoly profit function; optimum $p_1 = 55$ with $\pi_1 = 2,025$.

(c) $p_2 = 10$: undercutting produces losses; $p_1 = 10$ splits the market at zero margin; $p_1 > 10$ sells nothing. Maximum profit is 0.

(d) $10 < p_2 < p^M$: profit rises in p_1 up to just below p_2 — firm 1 wants to undercut marginally ($p_1 = p_2 - \epsilon$); at $p_1 = p_2$ profit drops to half, above it to zero.

6.3 Reaction function of firm 1

$$P_1(p_2) = 55 \text{ if } p_2 > 55; \quad p_2 - \epsilon \text{ if } 10 < p_2 \leq 55; \quad p_1 = 10 \text{ if } p_2 \leq 10$$

6.4 Nash equilibrium

Mutual undercutting drives prices to marginal cost: $p^A B = 10$, total quantity $x^A B = 90$ (45 per firm). No firm can profitably deviate: undercutting means losses, raising price means zero sales.

6.5 Surpluses at the equilibrium

$$PS = (10 - 10) \times 90 = 0 \quad CS = \frac{1}{2} \times 90 \times (100 - 10) = 4,050$$

6.6 Pareto efficiency

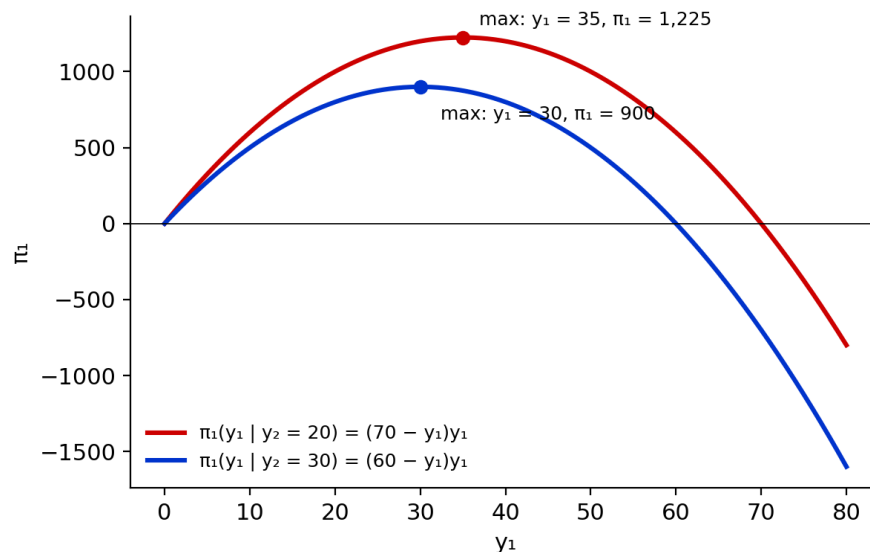
Yes. Price equals marginal cost, so the allocation replicates perfect competition: every unit valued above cost is produced and total surplus (4,050) is maximal — there is no deadweight loss. The firms would prefer to collude, but any such move would hurt consumers by more than it benefits firms, so the Bertrand outcome cannot be Pareto-improved. This is the 'Bertrand paradox': two firms suffice for the competitive outcome.

Problem 7

Cournot duopoly with inverse demand $P(y) = 100 - y$, where $y = y_1 + y_2$, and costs $C_i(y_i) = c \cdot y_i$ with $c = 10$.

7.1 Profit function of firm 1 for $y_2 = 20$ and $y_2 = 30$

$$\pi_1(y_1 | y_2) = (100 - y_1 - y_2 - 10) \cdot y_1 = (90 - y_2 - y_1) \cdot y_1$$



Each is a downward-opening parabola. The larger the rival's output, the lower and further left firm 1's profit hill: the residual demand shrinks, so both the optimal quantity ($35 \rightarrow 30$) and the attainable profit ($1,225 \rightarrow 900$) fall. Firm 1's best choice depends on what firm 2 does — the essence of strategic interdependence.

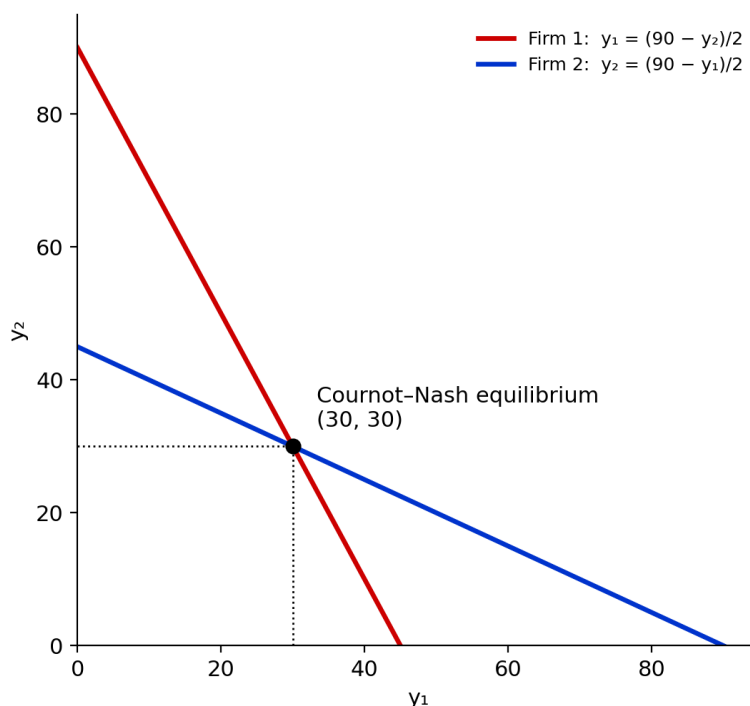
7.2 Reaction function of firm 1

Maximizing π_1 over y_1 gives the first-order condition $90 - y_2 - 2y_1 = 0$:

$$y_1^*(y_2) = (90 - y_2)/2$$

For $y_2 = 0$: $y_1 = 45$ — with no rival, firm 1 produces the monopoly quantity. For $y_2 = 90$: $y_1 = 0$ — the rival already supplies the competitive quantity, price equals marginal cost, and any additional unit would push price below cost. The reaction function spans the whole range between monopoly and exit.

7.3 Reaction functions and the Cournot-Nash equilibrium



By symmetry, $y_1 = y_2 = y$: $y = (90 - y)/2 \Rightarrow y_1^{CN} = y_2^{CN} = 30$, at the intersection of the two reaction functions.

7.4 Price and market supply

$$y^{CN} = 60 \quad p^{CN} = 100 - 60 = 40$$

7.5 Surpluses

$$PS = (40 - 10) \times 60 = 1,800 \text{ (900 per firm)} \quad CS = \frac{1}{2} \times 60 \times (100 - 40) = 1,800$$

7.6 Pareto efficiency

No. Price (40) exceeds marginal cost (10), so units valued above cost go unproduced. The competitive benchmark is $y = 90$, $p = 10$ with total surplus $\frac{1}{2} \times 90 \times 90 = 4,050$; the Cournot total is 3,600, leaving a deadweight loss of $\frac{1}{2} \times (40 - 10) \times (90 - 60) = 450$.

7.7 Moving along lines I and II from the Nash equilibrium (Fig. 15.1)

Starting at (30, 30), both firms can raise their profits by jointly reducing output — moving along the lines toward the collusive (cartel) allocation, where the two together produce the monopoly quantity of 45 (22.5 each) and share the monopoly profit of 2,025 instead of 1,800. What must be taken into consideration: this point is not a Nash equilibrium. Given the rival's reduced output, each firm's best response is to expand along its reaction function — every firm has a unilateral incentive to cheat. Sustaining the move therefore requires enforcement, e.g. repeated interaction with credible punishment strategies; a one-shot agreement will unravel.

7.8 General case with n identical firms

Firm i maximizes $(100 - Y - 10)y_i$, giving $90 - Y_{-i} - 2y_i = 0$. Imposing symmetry ($y_i = y$, $Y = ny$):

$$y_i^{CN} = 90/(n+1) \quad Y^{CN} = 90n/(n+1) \quad p^{CN} = 10 + 90/(n+1)$$

Check: $n = 1$ gives the monopoly (45, $p = 55$); $n = 2$ the duopoly above (30 each, $p = 40$).

7.9 Effect of the number of firms

As n rises, individual output $90/(n+1)$ falls, market output $90n/(n+1)$ rises toward the competitive quantity 90, and price $10 + 90/(n+1)$ falls toward marginal cost 10. Cournot competition thus spans the entire range from monopoly ($n = 1$) to perfect competition ($n \rightarrow \infty$): the more firms, the smaller each firm's market power and the smaller the deadweight loss.

Problem 8

8.1 When to use Bertrand vs. Cournot

Bertrand (price competition) fits markets where firms set prices and can readily serve whatever demand arrives — capacity is flexible and adjustment is fast: retail and online commerce, insurance, telecommunications tariffs. **Cournot** (quantity competition) fits markets where the key decision is capacity or production volume fixed in advance, with prices then clearing the market: raw materials and commodities (oil, cement, chemicals), electricity generation, airline seat capacity.

8.2 The two-stage model

In the two-stage model, firms first choose capacities (a long-run, hard-to-reverse decision) and then compete in prices given those capacities (a short-run, flexible decision). Solving backwards, the price stage is Bertrand-like but constrained by capacity; anticipating this, firms choose capacities strategically in stage one. The celebrated result (Kreps–Scheinkman) is that the equilibrium of this two-stage game replicates the Cournot outcome. The approach is interesting because it resolves the apparent contradiction between the two models — 'Cournot' is best read as a reduced form of capacity choice followed by price competition — and because it mirrors the actual sequencing of firms' decisions.

8.3 Why collude, and how can collusion be enforced?

Oligopolists jointly earn less than a monopolist would (e.g., $1,800 < 2,025$ in Problem 7): by restricting output or raising prices together toward the monopoly solution, all firms can gain — hence the incentive to collude. Because each firm can gain even more by secretly deviating, collusion must be made credible.

Since cartel contracts are illegal and unenforceable in court, enforcement must be self-sustaining: repeated interaction with trigger/punishment strategies (deviation is met with a lasting price war), sufficient patience (high weight on future profits), quick detection of deviations (market transparency), few and symmetric firms, and stable demand. Institutional devices such as information exchange or industry associations can facilitate — which is precisely why competition law targets them.

8.4 Is collusion stable and desirable?

Collusion is inherently fragile: the collusive point is not a Nash equilibrium of the stage game, so each member is permanently tempted to cheat, and entry by outsiders erodes the cartel's rents. The empirical evidence matches this: cartels form regularly and some persist for years (particularly with few members, homogeneous products, and good monitoring), but most eventually collapse through cheating, entry, demand shocks, or detection by authorities — often aided by leniency programs that make members defect. Collusion is not socially desirable: it moves the market toward the monopoly outcome, raising prices and creating deadweight loss while redistributing surplus from consumers to firms. That is why it is prohibited and prosecuted in virtually all competition-law regimes.

8.5 The three Cs of economics

The three Cs classify the fundamental problems that economic institutions must solve: **coordination** (aligning behavior when several outcomes are possible and actors must match expectations — e.g., standards, driving side, meeting points), **cooperation** (achieving jointly beneficial outcomes when each individual is tempted to free-ride or defect — the prisoner's-dilemma structure behind cartels, public goods, and common resources), and **commitment** (making promises or threats credible over time when one will later be tempted to renege — e.g., punishment threats sustaining collusion, or a regulator promising not to bail firms out).

The framework is helpful because seemingly unrelated economic and social problems share these underlying structures. Diagnosing which C is at stake immediately suggests the type of institution needed: focal points and conventions for coordination problems, monitoring and sanctions for cooperation problems, and binding rules, reputations, or delegation for commitment problems. Oligopoly theory illustrates all three at once — cartel members must coordinate on a price, cooperate against the temptation to cheat, and commit to credible punishments.